

Ling Chang

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EDUCATION

Rice University

Master in Computational Science and Engineering

Houston, TX

Aug. 2025 - Dec. 2026

- Course: Computational Math, Database System, Computer System and Architecture, Machine learning

The University of Texas at Dallas

Master of Science in Information Technology and Management

Dallas, TX

Aug. 2024 - Jul. 2025

- Course: Object-Oriented Programming, Advanced Statistics for Data Science, Causal Analytics, Big Data

National ChengChi University

Bachelor of Art in International Relations

Taipei, Taiwan

Aug. 2013 - Jul. 2017

- Course: Data Structures and Algorithm, Data Analytics and Application, Deep Learning, Research Methods

SKILLS

- **Programming Languages:** Python, SQL, NoSQL, C++, R, Javascript, HTML/CSS
- **Applications:** MongoDB, PostgreSQL, Hadoop/Spark, Shell Script, Docker, Git, Linux, CI/CD, Tableau, PowerBI, Flask, GCP, AWS, Azure, Hive, ETL, React.js, Flask, Neo4j, Model Context Protocol (MCP)
- **Methodologies:** Machine Learning (ML), Data Analytics, A/B Testing, Statistical Hypothesis Testing

SELECTED PROJECTS

KG2Prompt: Knowledge Graphs for LLM Prompt Construction

React.js, Flask, Neo4j, Large Language Models (LLMs), MCP

Jun. 2025

- Extended a biomedical KG-based retrieval system to support dynamic LLM prompt refinement with LLM agents and search tools deployed on MCP servers, reducing irrelevant prompt tokens by 35%.
- Developed React.js and Flask APIs to support responsive KG visualization and KG-guided prompt construction, reducing navigation time by 30%.

Credit Card Approval Modeling and Prediction

R, Logistic Regression, Decision Tree, Shiny

May 2025

- Trained interpretable ML models achieving over 90% accuracy, minimizing false approvals.
- Visualized feature importance and model performance using ROC curves, confusion matrices, etc.
- Built an interactive Shiny app to simulate applicant profiles and provide real-time approval predictions.

Flight Reservation Database System Design

MySQL, PostgreSQL

Dec. 2024

- Designed entity relationship diagram with data manipulation language (DML) queries and executed business analytics using joins, subqueries, and aggregate functions.
- Deployed normalized data definition language (DDL) scripts in third normal form (3NF), improving query efficiency by 30% and reducing redundancy by 20%.

PUBLICATION

[\[NeurIPS' 25 Workshop\]](#) Y.N. Chuang, L. Yu, G. Wang, L. Zhang, **L. Chang**, et. al., and X. Hu. "Confident or Seek Stronger: Exploring Uncertainty-Based On-device LLM Routing"

EXPERIENCE

Border Affairs Corps, National Immigration Agency

Senior Immigration Officer

Taipei, Taiwan

Feb. 2019 - Aug. 2024

- Led the team and Analyzed over 1,000 case data entries for offline and online testing, establishing a ML system with anomaly detection algorithms to improve criminal identification accuracy by 42%.